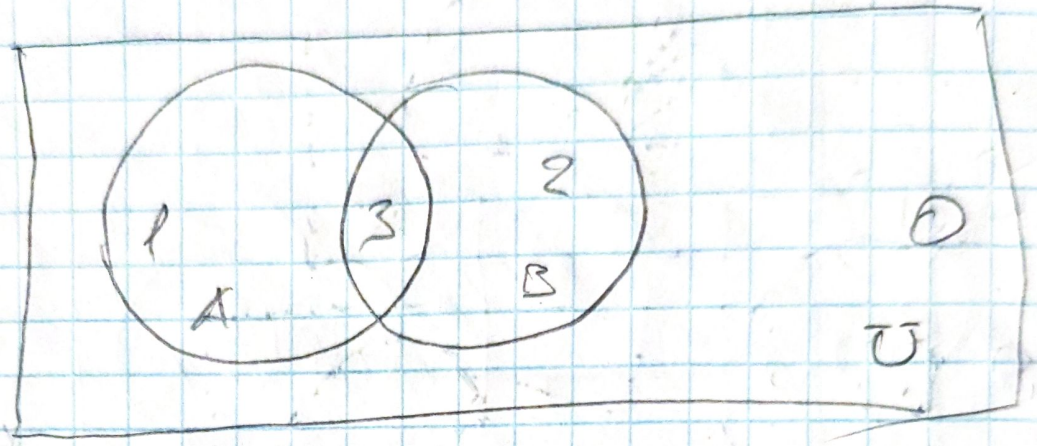
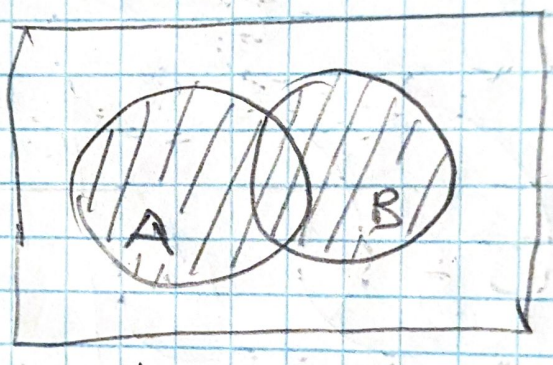
Блок А

А 1

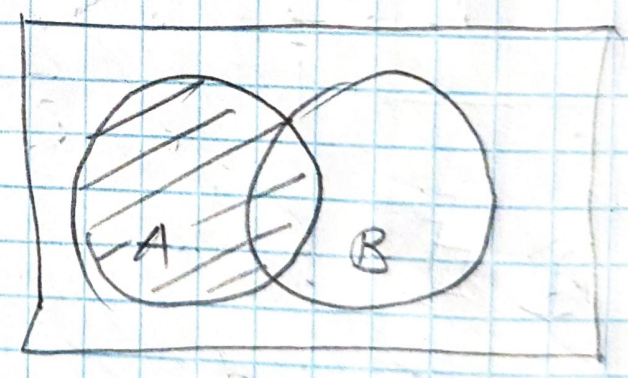
$A \cap (A \cup B) = A$



$A \cup B$



$A \cap (A \cup B)$



A

$A = \{1, 3\}$, $B = \{2, 3\}$

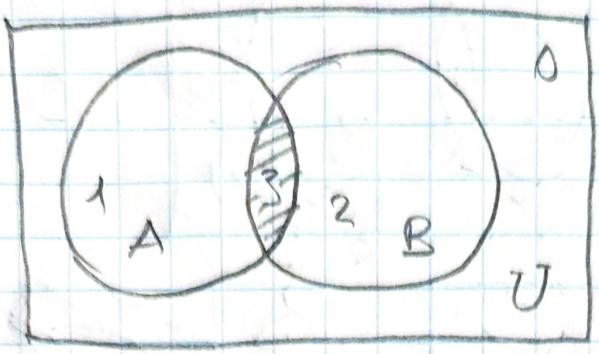
$A \cup B = \{1, 2, 3\}$,

$A \cap (A \cup B) = \{1, 3\} = A$

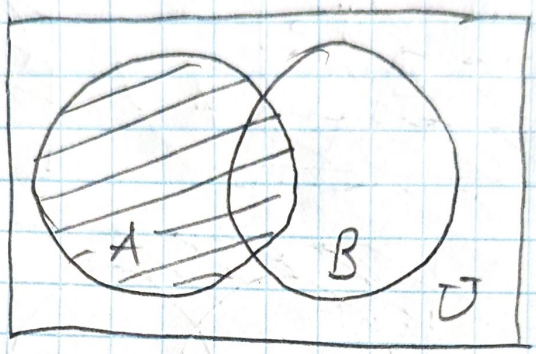
(A2)

$A \cup (A \cap B) = A$

$A \cap B$



$A \cup (A \cap B)$



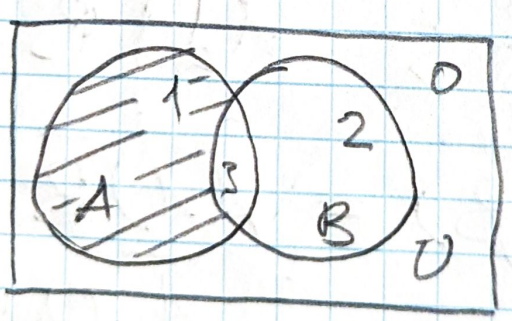
A

$A = \{1, 3\}$, $B = \{2, 3\}$, $A \cap B = \{3\}$

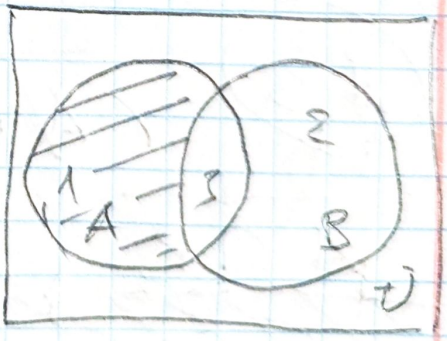
$A \cup (A \cap B) = \{1, 3\} = A$

(A3) $A \setminus B = A \cap \bar{B}$

$A \setminus B$



$A \cap \bar{B}$

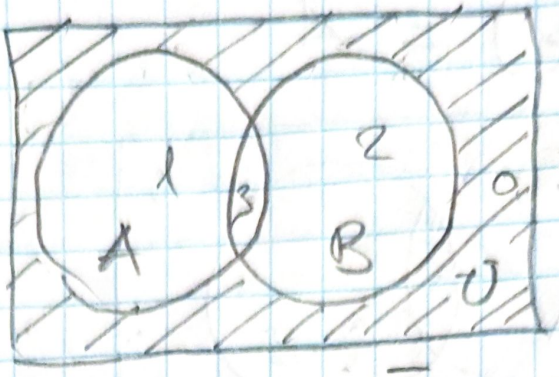


$A \setminus B = \{1\}$

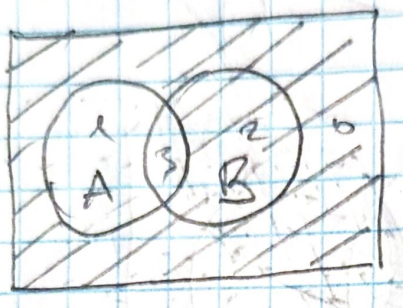
$A \cap \bar{B} = \{1\}$

A 4) $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$

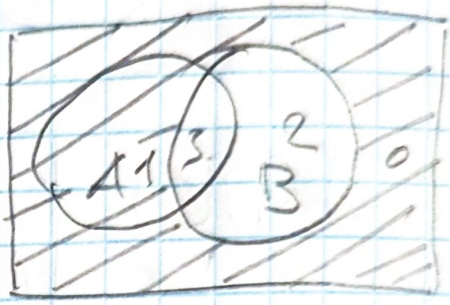
$\overline{A \cup B}$



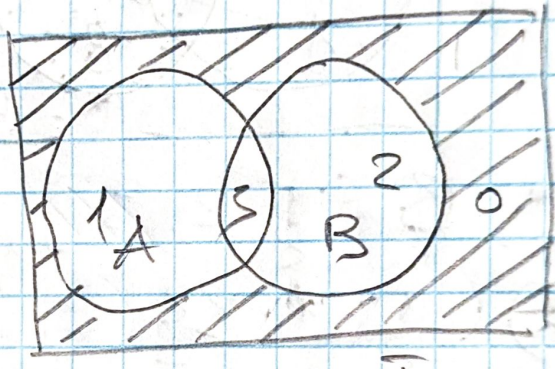
$\bar{A}$



$\bar{B}$



$\bar{A} \cap \bar{B}$

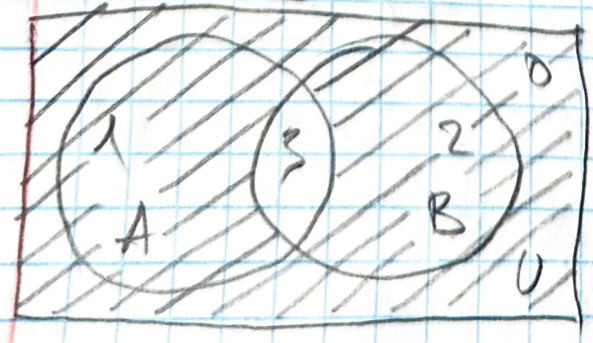


$\overline{A \cup B} = \{0\}$, $\bar{A} = \{0, 2\}$

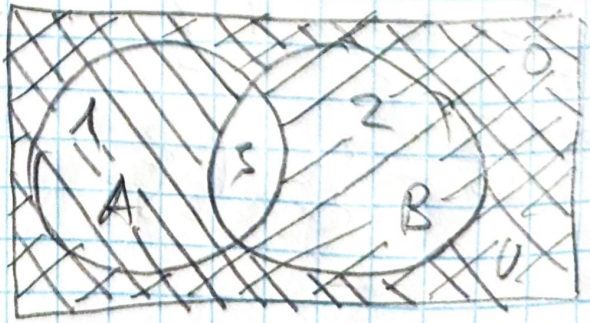
$\bar{B} = \{0, 1\}$, $\bar{A} \cap \bar{B} = \{0\}$

A 5) $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$

$\overline{A \cap B}$



$\bar{A} \cup \bar{B}$



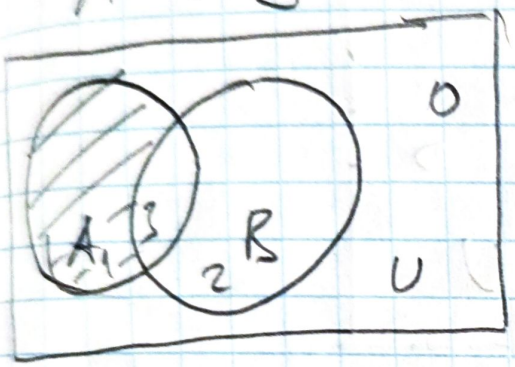
$A \cap B = \{3\}$ $\overline{A \cap B} = \{0, 1, 2\}$

$\bar{A} = \{0, 2\}$; $\bar{B} = \{0, 1\}$

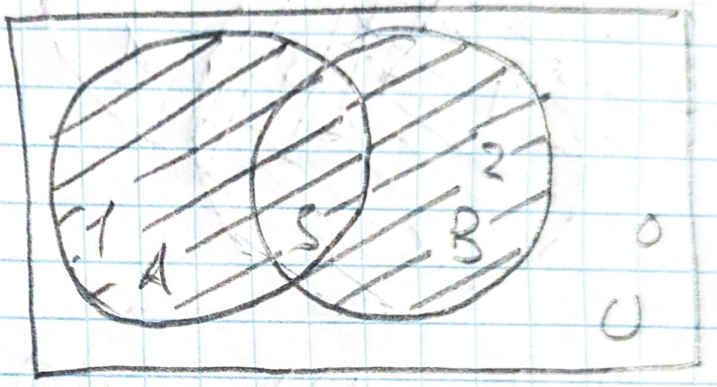
$\bar{A} \cup \bar{B} = \{0, 1, 2\} = \overline{A \cup B}$

(A6) $(A \setminus B) \cup B = A \cup B$

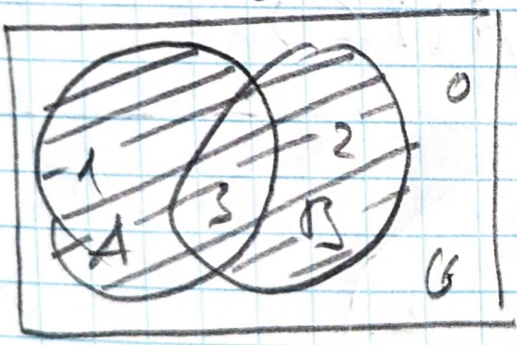
$A \setminus B$



$(A \setminus B) \cup B$



$A \cup B$



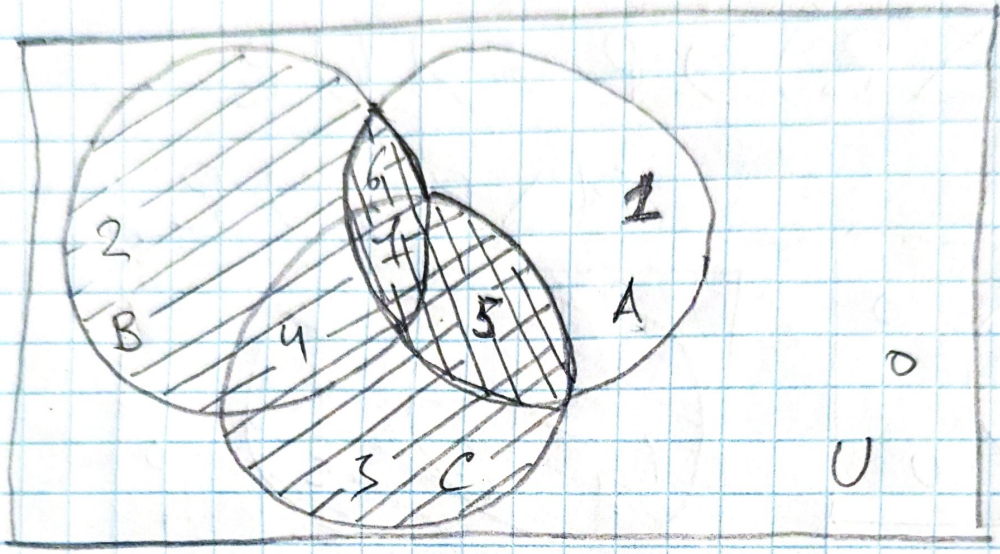
$A \setminus B = \{1\}$; $(A \setminus B) \cup B = \{1, 2, 3\}$

$A \cup B = \{1, 2, 3\}$

A 1

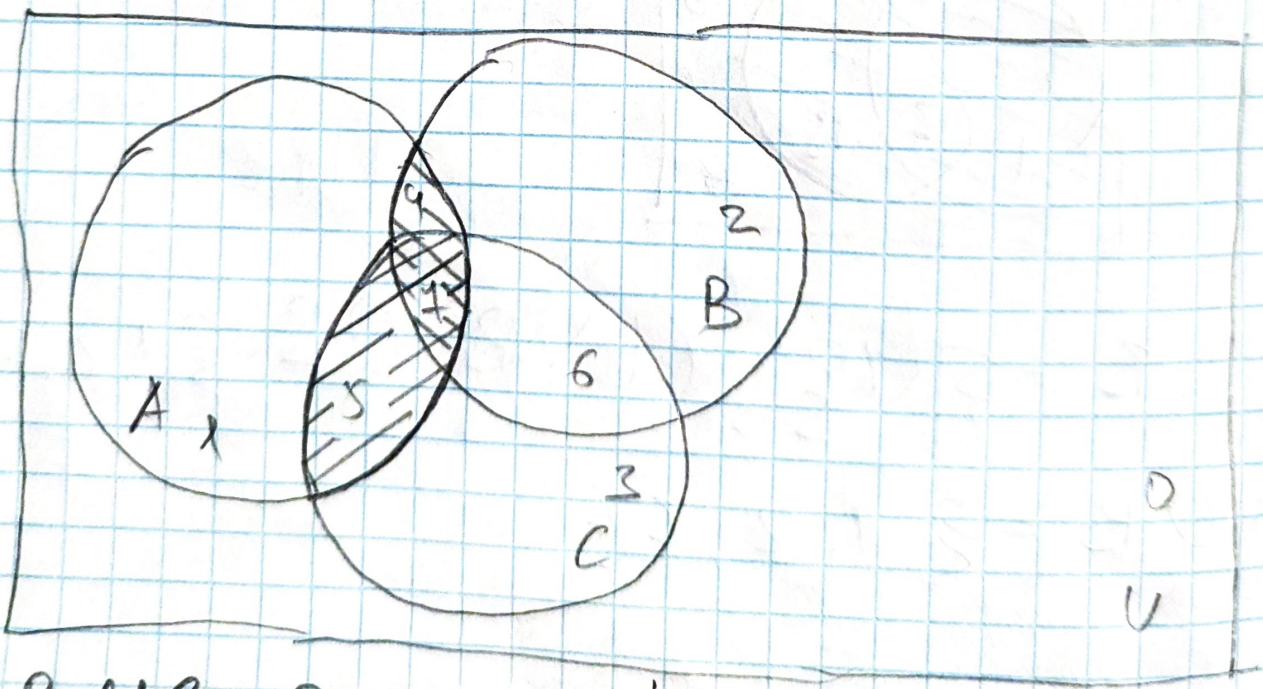
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$B \cup C$ - /// $A \cap (B \cup C)$ - \\\



$A \cap B$ - \\\ $A \cap C$ - ///

$(A \cap B) \cup (A \cap C)$ -



$B \cup C = \{2, 3, 4, 5, 6, 7\}$

$A \cap (B \cup C) = \{6, 7, 5\}$

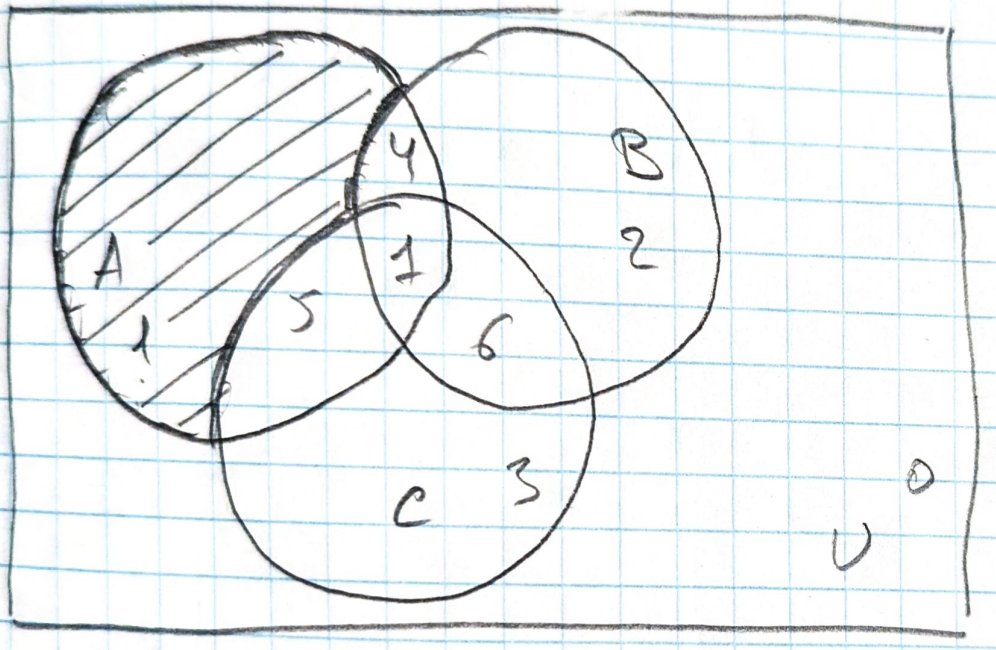
$A \cap B = \{4, 7\}$; $A \cap C = \{5, 7\}$

$(A \cap B) \cup (A \cap C) = \{4, 5, 7\}$

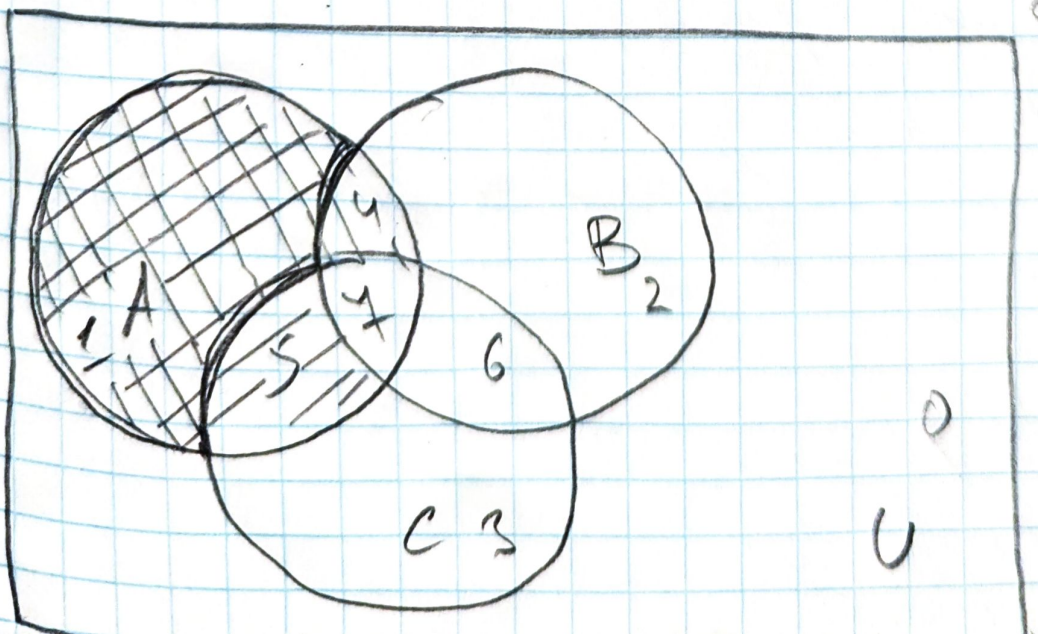
A8

$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$

$A \setminus (B \cup C) = \{1\}$



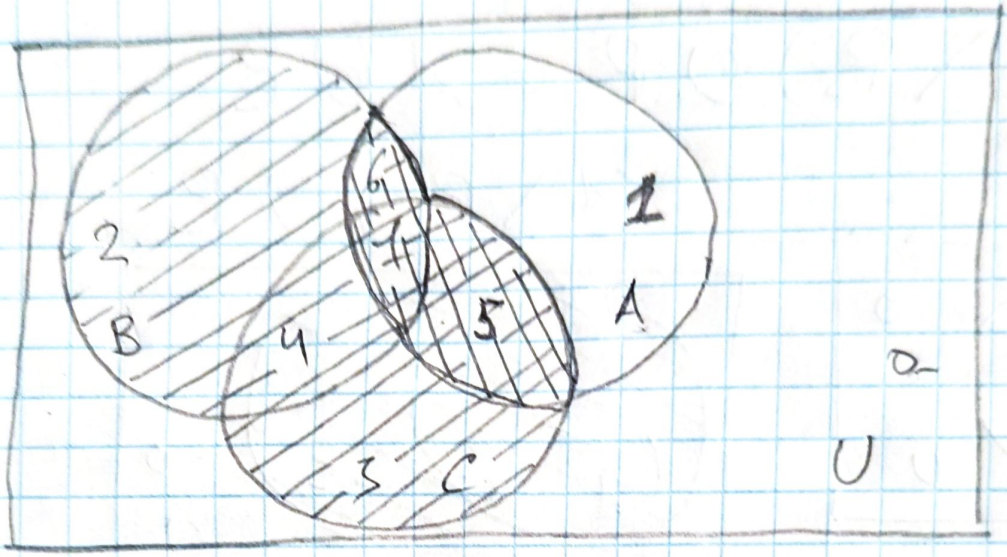
$(A \setminus B) \cap (A \setminus C) = \{1\}$



A 1

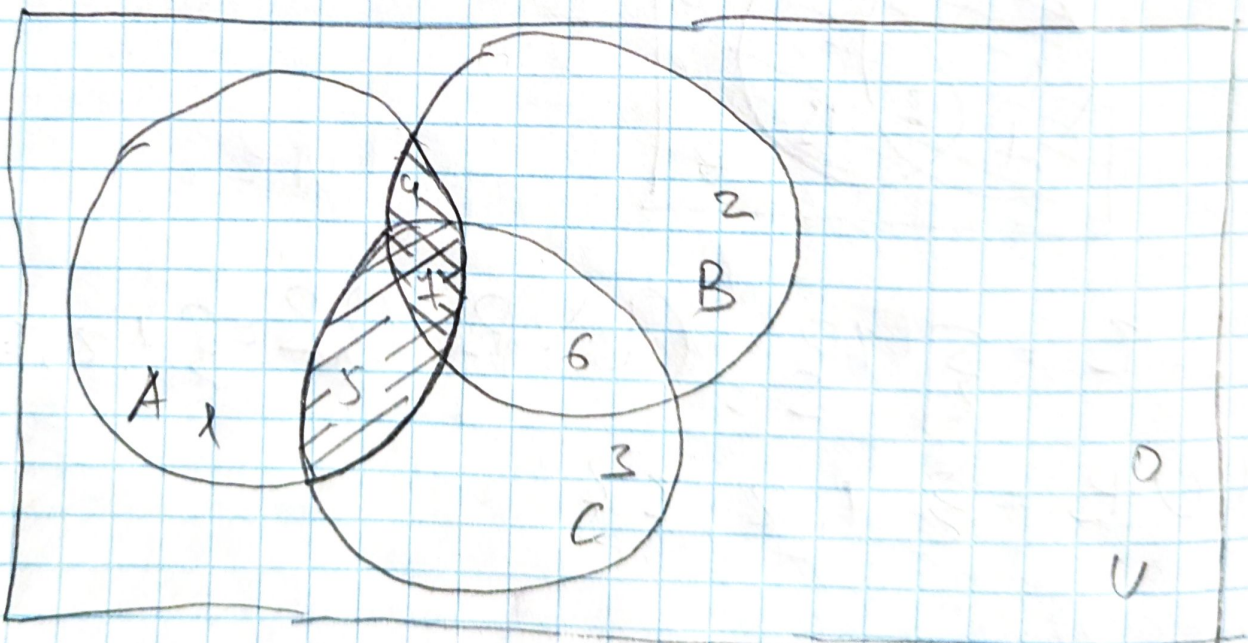
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$B \cup C$ - /// $A \cap (B \cup C)$ - \\\



$A \cap B$ - \\\ $A \cap C$ - ///

$(A \cap B) \cup (A \cap C)$ -



$B \cup C = \{2, 3, 4, 5, 6, 7\}$

$A \cap (B \cup C) = \{4, 5, 7\}$

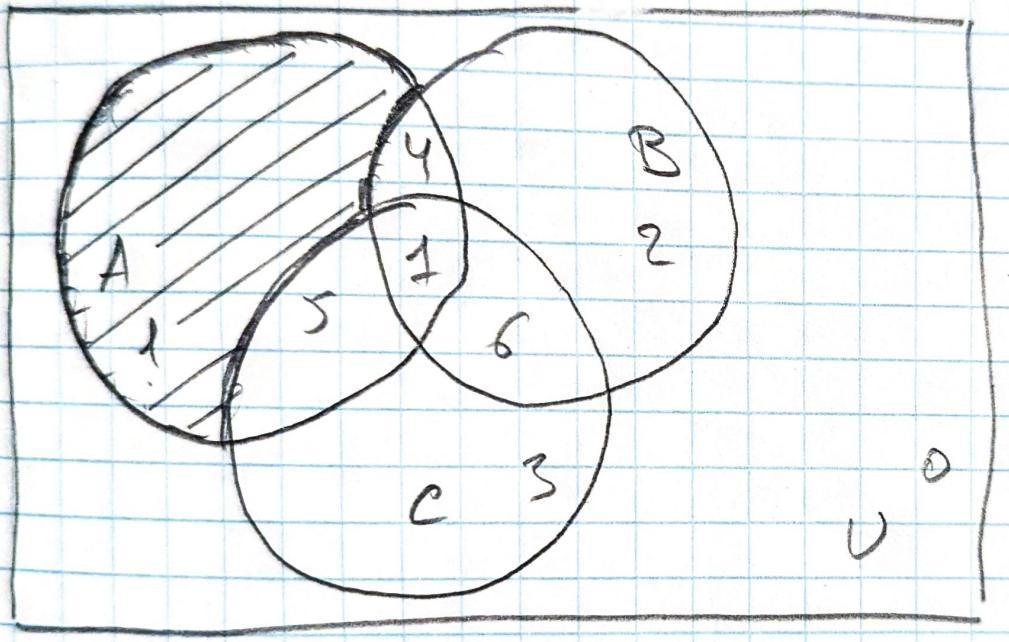
$A \cap B = \{4, 7\}$; $A \cap C = \{5, 7\}$

$(A \cap B) \cup (A \cap C) = \{4, 5, 7\}$

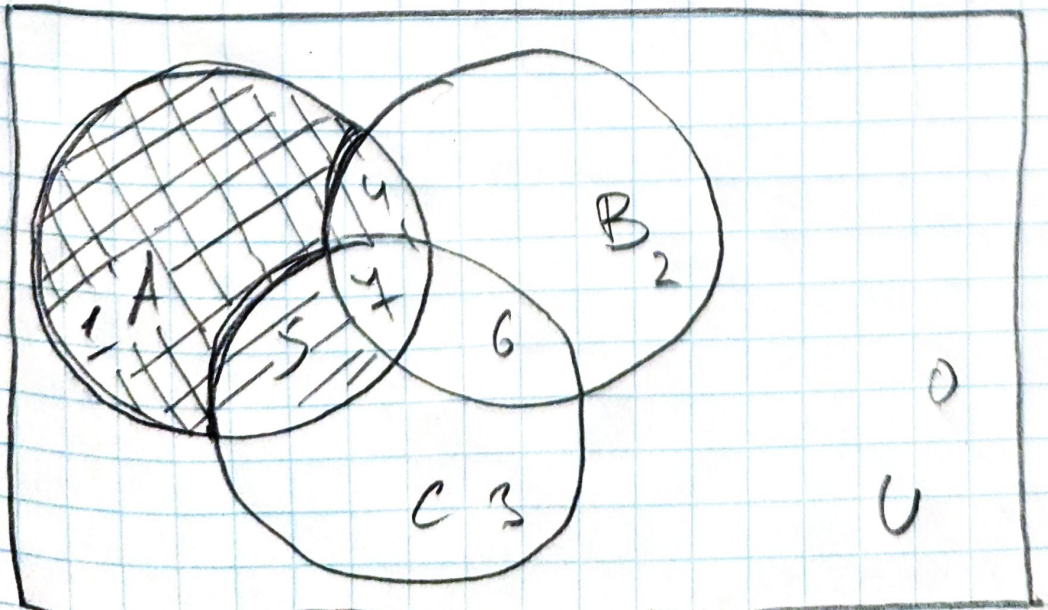
(A8)

$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$

$A \setminus (B \cup C) = \{1\}$



$(A \setminus B) \cap (A \setminus C) = \{1\}$



Блок В

(В1) $A \cup (A \cap B) = A$

1) дистрибутивность

$(A \cup A) \cap (A \cup B)$

2) идемпотентность

$A \cup A = A$

$A \cap (A \cup B)$ — поглощение

3) дистрибутивность

$(A \cap A) \cup (A \cap B)$

4) идемпотентность

$A \cap A = A$

$A \cup (A \cap B)$

5) закон поглощения

$A \cup (A \cap B) = A$

(B2) $A \setminus B = A \cap \overline{B}$

это формальное множество, которое можно разделить через эл. и Венна

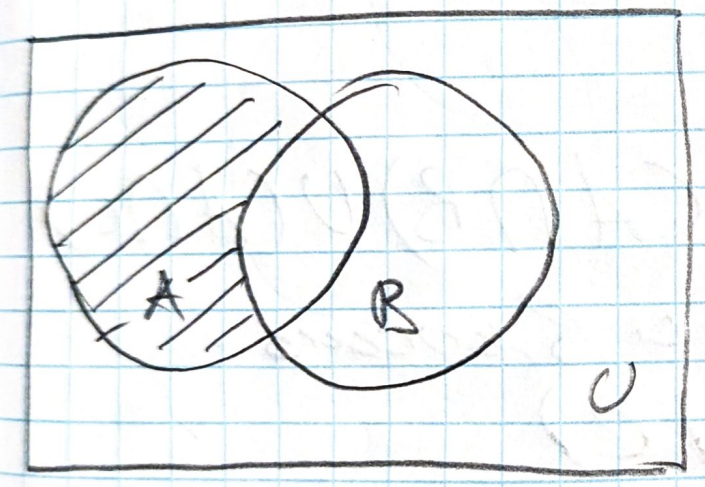
$\Rightarrow$ пусть $x \in A \setminus B$

$x \in A \wedge x \notin B \Rightarrow A \cap \overline{B}$

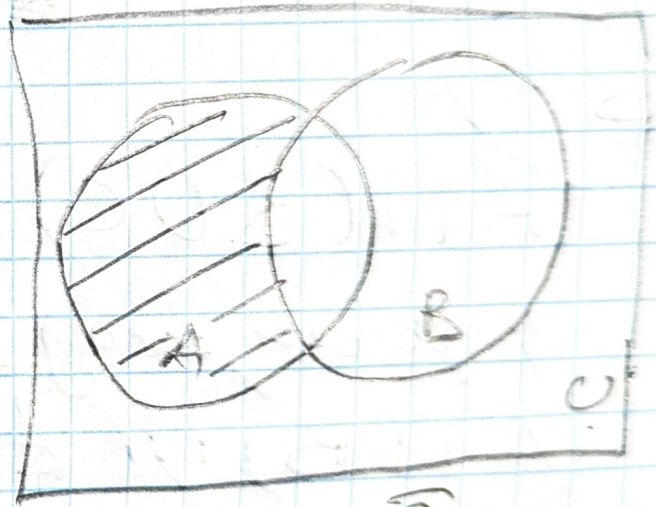
$\Leftarrow$ пусть $x \in A \cap \overline{B}$

$x \in A \wedge x \notin B$ по определению

зам $A \setminus B$



$A \setminus B$



$A \wedge \overline{B}$

$\equiv$

№3 $A \cup (\bar{A} \cap B) = A \cup B$

1) дистрибутивность.
$(A \cup \bar{A}) \cap (A \cup B)$

2) закон дополнения
$A \cup \bar{A} = U$ универсум
$U \cap (A \cup B)$

3) закон тождества
$U \cap A = A$
$A \cup B$
≡

№4
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

1) дистрибутивность.
$(A \cap B) \cup (A \cap C)$
≡

(B1) $A \cup (A \cap B) = A$

1) Задача тождественна $A = A \cap U$

$(A \cap U) \cup (A \cap B)$

2) Дистрибутивность в обратную сторону

$A \cap (U \cup B)$

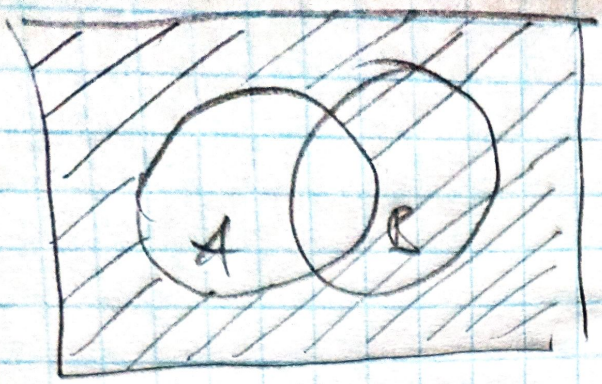
3) Задача решена

$U \cup B = U$

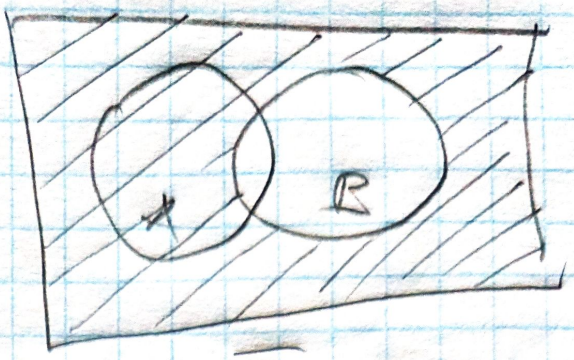
4) $A \cap U$ - задача тождества

$A \cap U = A$

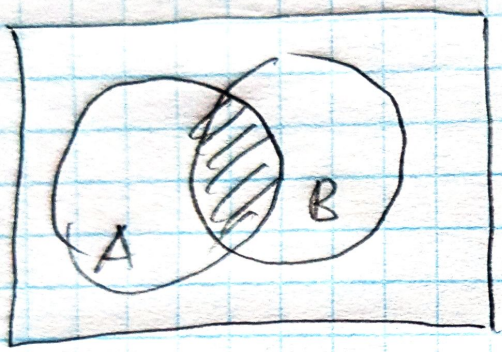
(B2) $x \in A \setminus B \Leftrightarrow x \in A \wedge x \notin B \Leftrightarrow x \in A \cap \overline{B}$



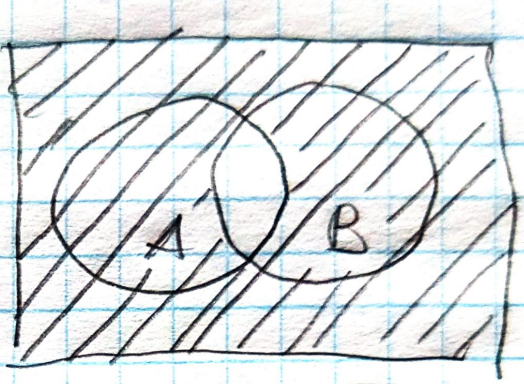
$\overline{A}$



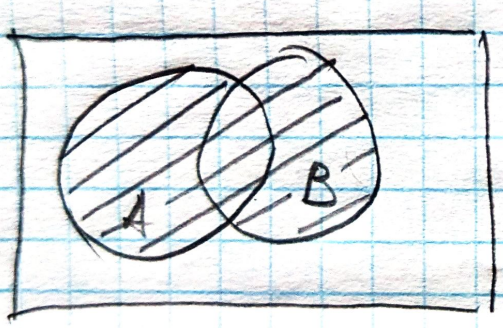
$\overline{B}$



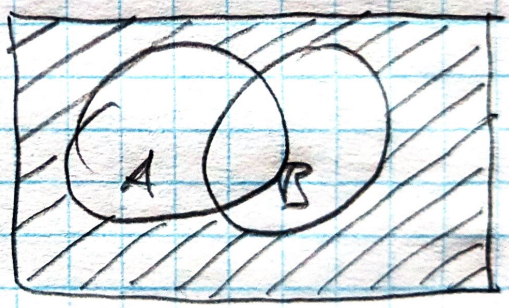
$A \cap B$



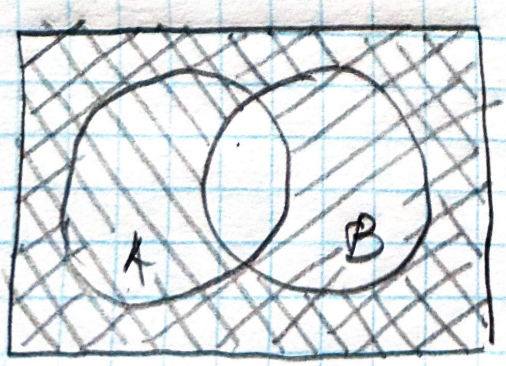
$\overline{A \cap B}$



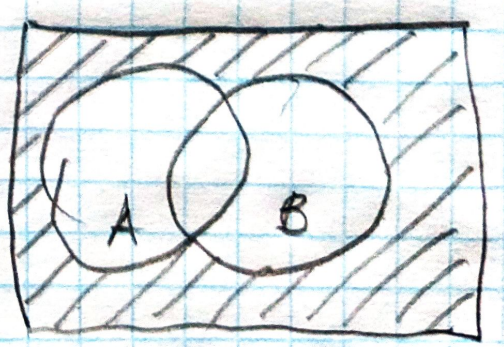
$A \cup B$



$\overline{A \cup B}$



$\overline{A} \cup \overline{B}$



$\overline{A} \cap \overline{B}$

$$A \setminus B = A \cap \overline{B}$$

$\Rightarrow$ пусть $x \in A \setminus B \Leftrightarrow$

$\Leftrightarrow x \in A \wedge x \notin B \Leftrightarrow$

$\Leftrightarrow x \in A \cap \overline{B}$

$\Leftarrow$ пусть $x \in A \cap \overline{B} \Leftrightarrow$

$\Leftrightarrow x \in A \wedge x \notin B \Leftrightarrow$

$\Leftrightarrow x \in A \setminus B$

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

$x \in A \setminus (B \cup C) \Leftrightarrow$

$\Leftrightarrow (x \in A) \wedge \neg (x \in B \cup C) \Leftrightarrow$

$\Leftrightarrow (x \in A) \wedge [\neg (x \in B) \wedge \neg (x \in C)] \Leftrightarrow$

$(x \in A) \Leftrightarrow (x \in A) \wedge (x \in A)$

~~$[(x \in A) \wedge (x \in A)]$~~

$[(x \in A) \wedge \neg (x \in B)] \wedge [(x \in A) \wedge \neg (x \in C)] \Rightarrow$

$\Rightarrow (x \in A \setminus B) \wedge (x \in A \setminus C) \Rightarrow$

$\Rightarrow (A \setminus B) \cap (A \setminus C)$